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Power efficient complementary amplifier and method thereof

Abstract

A method operates by receiving an input signal; transforming the input signal into a first transformed signal of a common-mode voltage equal to a first bias voltage; transforming the input signal into a second transformed signal of a common-mode voltage equal to a second bias voltage; detecting a peak of the first transformed signal and making the first bias voltage equal to a sum of a first DC (direct current) voltage and a first dynamic voltage; detecting a valley of the second transformed signal and making the second bias voltage equal to a sum of a second DC voltage and a second dynamic voltage; amplifying the first transformed signal into a first output signal across a first load using a N-type cascode amplifier; and amplifying the second transformed signal into a second output signal across a second load using a P-type cascode amplifier.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
9964985	12/2017	Leong	N/A	N/A
10447218	12/2018	Xie	N/A	N/A

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Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

(1) The present invention generally relates to amplifier and particularly to power efficient complementary amplifier.

Description of Related Art

(2) A conventional common-source amplifier uses a NMOST (n-channel metal oxide semiconductor field-effect transistor) to fulfill an amplification function; the NMOST has a source (terminal), a gate (terminal), and a drain (terminal), wherein an input voltage is received at the gate, a ground node is connected to the source, and an output current is delivered from a drain to a load that establishes an output voltage in response to the output current in accordance with an impedance of the load. Ideally, an incremental change of the input voltage will result in a proportional incremental change of the output voltage, wherein a ratio of incremental changes between the output voltage and the input voltage is referred to as a gain. A linearity of the amplification function is gauged by how well the gain can sustain and maintain approximately the same value when an amplitude of the input voltage increases, and it depends on a bias condition. A conventional common-source amplifier biased in a deeper class-A region, wherein a quiescent bias current is higher, can have a better linearity, at a cost of a poorer power efficiency. To preserve a good power efficiency while maintaining a good linearity, a dynamic bias scheme is often used, wherein a bias condition is dynamically adjusted in accordance with an amplitude of the input voltage in a way that the common-source amplifier is biased in a deeper class-A region in the presence of an input voltage of a larger amplitude. A conventional common-source amplifier, however, often encounters issues of ground bounce and source degeneration due to a nonzero resistance of a physical connection between a source and a ground node. Concepts of “common-source amplifier,” “bias,” “class-A region,” “power efficiency,” “source degeneration,” and “ground bounce” are well understood by those of ordinary skills in the art and thus not described in detail here.

(3) In U.S. Pat. No. 10,447,218– B1, Xie, Leong, and Lin disclosed a complementary amplifier that uses both a NMOST and a PMOST (p-channel metal oxide semiconductor field-effect transistor) to form a hybrid differential pair that share a common source node, i.e., a source of the NMOST is directly connected to a source of the PMOST, so that the issues of ground bounce and source degeneration that conventional common-source amplifiers often encounter can be mitigated. Due to a difference in amplifier topology, however, a dynamic bias scheme used in a conventional common-source amplifier is no longer applicable to a complementary amplifier.

(4) What is desired is a dynamic bias scheme for a complementary amplifier.

BRIEF SUMMARY OF THIS INVENTION

(5) In an embodiment, an amplifier comprises: a transformer configured to transform an input signal received from a primary coil into a first transformed signal of a first bias voltage and a second transformed signal of a second bias voltage via a first secondary coil and a secondary coil, respectively; a N-type dynamic bias circuit configured to receive the first transformed signal and output the first bias voltage in accordance with a first DC (direct current) voltage and a peak of the first transformed signal; a P-type dynamic bias circuit configured to receive the second transformed signal and output the second bias voltage in accordance with a second DC voltage and a valley of the second transformed signal; a N-type cascode amplifier configured to receive the first transformed signal and output a first output signal across a first load; and a P-type cascode amplifier configured to receive the second transformed signal and output a second output signal across a second load, wherein: the N-type cascode amplifier and the P-type cascode amplifier share a common source node.

(6) In an embodiment, a method comprises: receiving an input signal; transforming the input signal into a first transformed signal of a common-mode voltage equal to a first bias voltage; transforming the input signal into a second transformed signal of a common-mode voltage equal to a second bias voltage; detecting a peak of the first transformed signal and making the first bias voltage equal to a sum of a first DC (direct current) voltage and a first dynamic voltage approximately proportional to the peak; detecting a valley of the second transformed signal and making the second bias voltage equal to a sum of a second DC voltage and a second dynamic voltage approximately proportional to the valley; amplifying the first transformed signal into a first output signal across a first load using a N-type cascode amplifier; and amplifying the second transformed signal into a second output signal across a second load using a P-type cascode amplifier, wherein the N-type cascode amplifier and the P-type amplifier share a common source node.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 shows a schematic diagram of an amplifier in accordance with an embodiment of the present invention.

(2) FIG. 2 shows an exemplary waveform of an input signal.

(3) FIG. 3 shows a schematic diagram of a N-type dynamic bias circuit in accordance with an embodiment of the present disclosure.

(4) FIG. 4 shows a schematic diagram of a P-type dynamic bias circuit in accordance with an embodiment of the present disclosure.

(5) FIG. 5 shows a flow diagram of a method in accordance with the present invention.

DETAILED DESCRIPTION OF THIS INVENTION

(6) The present invention relates to amplifiers. While the specification describes several example embodiments of the invention considered favorable modes of practicing the invention, it should be understood that the invention can be implemented in many ways and is not limited to the particular examples described below or to the particular manner in which any features of such examples are

implemented. In other instances, well-known details are not shown or described to avoid obscuring aspects of the invention.

(7) Persons of ordinary skill in the art understand terms and basic concepts related to microelectronics that are used in this disclosure, such as “voltage,” “current,” “signal,” “differential signal,” “common mode,” “capacitor,” “inductor,” “resistor,” “transistor,” “MOST (metal-oxide semiconductor field-effect transistor),” “PMOST (p-channel metal oxide semiconductor field-effect transistor),” “NMOST (n-channel metal oxide semiconductor field-effect transistor),” “AC (alternating current),” “DC (direct current),” “DC coupled,” “AC coupled,” “source,” “gate,” “drain,” “node,” “ground node,” “power supply node,” “cascode,” “common-source amplifier,” “common-gate amplifier,” “load,” “impedance,” and “cascode amplifier.” Those of ordinary skill in the art can also readily recognize a symbol of a MOST (metal-oxide semiconductor field effect transistor), and its associated “source,” “gate,” and “drain” terminals. Terms and basic concepts like these are understood by those of ordinary skill in the art and thus will not be explained in detail here.

(8) Persons of ordinary skill in the art can recognize a resistor symbol, a capacitor symbol, an inductor symbol, and a MOST (metal-oxide semiconductor field effect transistor) symbol, for both PMOST (p-channel metal-oxide semiconductor field effect transistor) and NMOST (n-channel metal-oxide semiconductor field effect transistor), and can identify a “source” terminal, a “gate” terminal, and a “drain” terminal of a MOST. For brevity, in this present disclosure, in a context of referring to a MOST, a “source terminal” is simply referred to as “source,” a “gate terminal” is simply referred to as “gate,” and a “drain terminal” is simply referred to as “drain.”

(9) Those of ordinary skill in the art can read schematics of a circuit comprising resistors, capacitors, inductors, NMOST, and PMOST, and do not need a verbose description about how one transistor, resistor, inductor, or capacitor connects with another in the schematics.

(10) Throughout this disclosure, “DC” stands for direct current, and “AC” stands for alternating current. A DC node is a node of a substantially fixed electric potential. In particular, “V.sub.DD” denotes a first DC node referred to as a power node, and “V.sub.SS” denotes a second DC node referred to as a ground node. A DC voltage refers to a substantially stationary voltage. An AC voltage refers to a voltage that varies with time in an oscillatory manner and is also referred to as a dynamic voltage.

(11) A common-source amplifier is embodied by a MOST configured to receive an input voltage form a gate and output an output current via a drain, wherein a source is connected to a DC node. A common-source amplifier embodied by a NMOST (PMOST) is referred to an N-type (P-type) common-source amplifier.

(12) A common-gate amplifier is embodied by a MOST configured to receive an input current from a source and output an output current via a drain, wherein a gate is connected to a DC node. A common-gate amplifier embodied by a NMOST (PMOST) is referred to as an N-type (P-type) common-gate amplifier.

(13) A cascode amplifier is a cascade of a common-source amplifier and a common-gate amplifier of the same type, wherein an output current of the common-source amplifier is an input current of the common-gate amplifier. A cascode amplifier embodied by NMOST (PMOST) is referred to as a N-type (P-type) cascode amplifier.

(14) A source follower is embodied by a MOST configured to receive an input voltage at a gate and output an output voltage at a source, wherein a drain connects to a DC node or a circuit node of a sufficiently low impedance. A source follower embodied by a NMOST (PMOST) is referred to as a N-type (P-type) source follower.

(15) A circuit is a collection of a transistor, a capacitor, an inductor, a resistor, and/or other electronic devices inter-connected in a certain manner to embody a certain function. A network is a circuit or a collection of circuits configured to embody a certain function.

(16) In this present disclosure, a “circuit node” is simply referred to as a “node” for short, as the

meaning is clear from a context of microelectronics won't cause confusion.

(17) In this present disclosure, a signal is a voltage of a variable level that can vary with time. A (voltage) level of a signal at a moment represents a state of the signal at that moment. A differential signal, or a signal in a differential embodiment, comprises a first voltage denoted with a subscript appended with “+” and a second voltage denoted with a subscript appended with “-.” For instance, signal V.sub.i (V.sub.in, V.sub.ip, V.sub.op, V.sub.on) comprises V.sub.i+ (V.sub.in+, V.sub.ip+, V.sub.op+, V.sub.on+) and V.sub.i- (V.sub.in-, V.sub.ip-, V.sub.op-, V.sub.on-). A difference in level between the first voltage and the second voltage represents a state of the differential signal.

(18) A schematic diagram of an amplifier **100** in accordance with an embodiment of the present disclosure is shown in FIG. **1**. Amplifier **100** comprises: a trifilar transformer **150** configured to receive an input signal V.sub.i (which comprises two voltages V.sub.i+ and V.sub.i- in a differential embodiment) and output a first transformed signal V.sub.in (which comprises two voltages V.sub.in+ and V.sub.in- in a differential embodiment) and a second transformed signal V.sub.ip (which comprises two voltages V.sub.ip+ and V.sub.ip- in a differential embodiment) in accordance with a first bias voltage V.sub.b1 and a second bias voltage V.sub.b2, respectively; a N-type dynamic bias circuit **160** configured to receive the first transformed signal V.sub.in (comprising V.sub.in+ and V.sub.in-) and output the first bias voltage V.sub.b1 in accordance with a first DC (direct current) voltage V.sub.dc1; a P-type dynamic bias circuit **170** configured to receive the second transformed signal V.sub.ip (comprising V.sub.ip+ and V.sub.ip-) and output the second bias voltage V.sub.b2 in accordance with a second DC voltage V.sub.dc2; a N-type cascode amplifier **110** configured to receive the first transformed signal V.sub.in (comprising V.sub.in+ and V.sub.in-) and output a first output signal V.sub.on (which comprises two voltages V.sub.on+ and V.sub.on- in a differential embodiment) across a first load **130** in accordance with a third bias voltage V.sub.b3; and a P-type cascode amplifier **120** configured to receive the second transformed signal V.sub.ip (comprising V.sub.ip+ and V.sub.ip-) and output a second output signal V.sub.op (which comprises two voltages V.sub.op+ and V.sub.op- in a differential embodiment) across a second load **140** in accordance with a fourth bias voltage V.sub.b4. As mentioned earlier, throughout this disclosure, “V.sub.DD” denotes a power supply node, and “V.sub.SS” denotes a ground node.

(19) Trifilar transformer **150** comprises a primary coil **151**, a first secondary coil **152**, and a second secondary coil **153**. A first magnetic coupling K.sub.n between the primary coil **151** and the first secondary coil **152** transforms V.sub.i+ and V.sub.i- into V.sub.in+ and V.sub.in-, wherein a common-mode voltage of V.sub.in+ and V.sub.in- is equal to V.sub.b1 that is provided at a center-tap of the first secondary coil **152**. A second magnetic coupling K.sub.p between the primary coil **151** and the second secondary coil **153** transforms V.sub.i+ and V.sub.i- into V.sub.ip+ and V.sub.ip-, wherein a common-mode voltage of V.sub.ip+ and V.sub.ip- is equal to V.sub.b2 that is provided at a center-tap of the second secondary coil **153**.

(20) In an embodiment, the input signal V.sub.i (comprising V.sub.i+ and V.sub.i-) is a RF (radio frequency) signal of an amplitude that may vary with time. An exemplary waveform of V.sub.i+ and V.sub.i- is shown in FIG. **2**. As shown, V.sub.i+ and V.sub.i- are of the same waveform but opposite in polarity, and both are oscillatory (i.e., sinusoidal) and bounded between a peak and a valley. Mathematically, V.sub.i+ and V.sub.i- can be modeled as follows:

$$(21) V_{i+} = A(t)\sin(\omega t + \varphi(t)) \quad (1) \quad V_{i-} = -A(t)\sin(\omega t + \varphi(t)) \quad (2)$$

(22) Here, ω denotes an angular frequency, t denotes a time variable, $A(t)$ denotes an amplitude function of time, $\varphi(t)$ denotes a phase function of time. Note that $A(t)$ and $-A(t)$ represent the peak and the valley of the input signal V.sub.i, respectively.

(23) V.sub.in+ and V.sub.in- can be modeled as follows:

$$(24) V_{in+} = V_{b1} + A(t)\sin(\omega t + \varphi(t) + \varphi_n(t)) \quad (3)$$

$$V_{in-} = V_{b1} - A(t)\sin(\omega t + \varphi(t) + \varphi_n(t)) \quad (4)$$

(25) Here, $\alpha_{\text{sub.n}}$ denotes a voltage-scaling factor depending on a turn ratio between the primary coil **151** and the first secondary coil **152**, and $\theta_{\text{sub.n}}(t)$ denotes a phase shift caused by the magnetic coupling $K_{\text{sub.n}}$ between the primary coil **151** and the first secondary coil **152**.

(26) Likewise, $V_{\text{sub.ip}+}$ and $V_{\text{sub.ip}-}$ can be modeled as follows:

$$(27) \quad V_{\text{ip}+} = V_{b2} + \alpha_{\text{sub.p}} A(t) \sin(\omega t + \theta_{\text{sub.p}}(t)) \quad (3)$$

$$V_{\text{ip}-} = V_{b2} - \alpha_{\text{sub.p}} A(t) \sin(\omega t + \theta_{\text{sub.p}}(t)) \quad (4)$$

(28) Here, $\alpha_{\text{sub.p}}$ denotes a voltage-scaling factor depending on a turn ratio between the primary coil **151** and the second secondary coil **153**, and $\theta_{\text{sub.p}}(t)$ denotes a phase shift caused by the magnetic coupling $K_{\text{sub.p}}$ between the primary coil **151** and the second secondary coil **153**.

(29) In an embodiment, $\alpha_{\text{sub.p}}$ is approximately equal to $\alpha_{\text{sub.n}}$, while $\theta_{\text{sub.n}}(t)$ is approximately the same as $\theta_{\text{sub.p}}(t)$.

(30) N-type dynamic bias circuit **160** detects a peak of $V_{\text{sub.in}+}$ and $V_{\text{sub.in}-}$ and outputs $V_{\text{sub.b1}}$ such that $V_{\text{sub.b1}}$ is equal to $V_{\text{sub.dc1}}$ plus a dynamical voltage that is proportional to $A(t)$, that is:

$$(31) \quad V_{b1} = V_{\text{dc1}} + \beta_{\text{sub.n}} A(t) \quad (5)$$

Here, $\beta_{\text{sub.n}}$ denotes a gain factor of the N-type dynamic bias circuit **160**.

(32) Likewise, P-type dynamic bias circuit **170** detects a valley of $V_{\text{sub.ip}+}$ and $V_{\text{sub.ip}-}$ and outputs $V_{\text{sub.b2}}$ such that $V_{\text{sub.b2}}$ is equal to $V_{\text{sub.dc2}}$ plus a dynamical voltage that is proportional to $-A(t)$, that is:

$$(33) \quad V_{b2} = V_{\text{dc2}} - \beta_{\text{sub.p}} A(t) \quad (6)$$

Here, $\beta_{\text{sub.p}}$ denotes a gain factor of the P-type dynamic bias circuit **170**.

(34) In an embodiment, $\beta_{\text{sub.n}}$ is approximately equal to $\beta_{\text{sub.p}}$.

(35) A schematic diagram of an N-type dynamic bias circuit **300** that can be used to embody the N-type dynamic bias circuit **160** is shown in FIG. 3. N-type dynamic bias circuit **300** comprises: two AC (alternating current) coupling capacitors **331** and **332**; two DC (direct current) coupling resistors **333** and **334**; a peak detector **310** comprising three NMOST **311**, **312**, and **313**; a replica peak detector comprising three NMOST **321**, **322**, and **323**; and an operational amplifier **330**. The peak detector **310** receives two voltages $V'_{\text{sub.in}+}$ and $V'_{\text{sub.in}-}$ that are AC coupled to $V_{\text{sub.in}+}$ and $V_{\text{sub.in}-}$ via the two AC coupling capacitors **331** and **332**, respectively, wherein a DC level of $V'_{\text{sub.in}+}$ and $V'_{\text{sub.in}-}$ is set to a gate voltage $V_{\text{sub.gn}}$ via the two DC coupling resistors **333** and **334**, respectively. NMOST **311** and **312** are both configured as a source follower that receive $V'_{\text{sub.in}+}$ and $V'_{\text{sub.in}-}$, respectively, and jointly output $V_{\text{sub.b1}}$ in accordance with a bias current provided by NMOST **313** that is configured as a current source controlled by a bias voltage $V_{\text{sub.on}}$. $V_{\text{sub.b1}}$ will follow whichever is higher between $V'_{\text{sub.in}+}$ and $V'_{\text{sub.in}-}$, thus fulfilling a peak detection function. The replica peak detector **320** has the same circuit topology as the peak detector **310**. NMOST **321** and **322** are both configured as a source follower that receive V_{an} and jointly output a replica DC voltage $V'_{\text{sub.dc1}}$ in accordance with a bias current provided by NMOST **323** that is configured as a current source controlled by the bias voltage $V_{\text{sub.bn}}$. The operational amplifier **330** outputs the gate voltage $V_{\text{sub.gn}}$ in accordance with a difference between $V_{\text{sub.dc1}}$ and $V'_{\text{sub.dc1}}$. The operational amplifier **330** and NMOST **321** and **322** form a negative feedback control loop to make $V'_{\text{sub.dc1}}$ approximately equal to $V_{\text{sub.dc1}}$. Since the replica peak detector **320** is a replica of the peak detector **310**, a DC level of $V_{\text{sub.b1}}$ will be approximately equal to $V'_{\text{sub.dc1}}$ and thus approximately equal to $V_{\text{sub.dc1}}$ when $V_{\text{sub.in}+}$ and $V_{\text{sub.in}-}$ are quiescent (i.e., $A(t)=0$). When $V_{\text{sub.in}+}$ and $V_{\text{sub.in}-}$ are dynamic (i.e., $A(t) \neq 0$), an AC component of $V_{\text{sub.in}+}$ and $V_{\text{sub.in}-}$ will be coupled to $V'_{\text{sub.in}+}$ and $V'_{\text{sub.in}-}$, and a peak is detected and added to $V_{\text{sub.b1}}$ on top of $V_{\text{sub.dc1}}$, thus fulfilling the function described by equation (5). Operational amplifiers are well known in the prior art and thus not described in detail here. The operational amplifier **330** can be implemented using a variety of circuits known in the

prior art at a discretion of circuit designer.

(36) A schematic diagram of a P-type dynamic bias circuit **400** that can be used to embody the P-type dynamic bias circuit **170** is shown in FIG. **4**. P-type dynamic bias circuit **400** comprises: two AC (alternating current) coupling capacitors **431** and **432**; two DC (direct current) coupling resistors **433** and **434**; a valley detector **410** comprising three PMOST **411**, **412**, and **413**; a replica valley detector **420** comprising three PMOST **421**, **422**, and **423**; and an operational amplifier **430**. The valley detector **410** receives two voltages $V'.sub.ip+$ and $V'.sub.ip-$ that are AC coupled to $V.sub.ip+$ and $V.sub.ip-$ via the two AC coupling capacitors **431** and **432**, respectively, wherein a DC (direct current) level of $V'.sub.ip+$ and $V'.sub.ip-$ is set to be equal to a gate voltage $V.sub.gp$ via the two DC coupling resistors **433** and **434**, respectively. PMOST **411** and **412** are both configured as a source follower that receive $V'.sub.ip+$ and $V'.sub.ip-$, respectively, and jointly output $V.sub.b2$ in accordance with a bias current provided by PMOST **413** that is configured as a current source controlled by a bias voltage $V.sub.bp$. $V.sub.b2$ will follow whichever is lower between $V'.sub.ip+$ and $V'.sub.ip-$, thus fulfilling a valley detection function. The replica valley detector **420** has the same circuit topology as the valley detector **410**. PMOST **421** and **422** are both configured as a source follower that receive $V.sub.gp$ and jointly output a replica DC voltage $V'.sub.dc2$ in accordance with a bias current provided by PMOST **423** that is configured as a current source controlled by the bias voltage $V.sub.bp$. The operational amplifier **430** outputs the gate voltage $V.sub.gp$ in accordance with a difference between $V.sub.dc2$ and $V'.sub.dc2$. The operational amplifier **430** and PMOST **421** and **422** form a negative feedback control loop to make $V'.sub.dc2$ approximately equal to $V.sub.dc2$. Since the replica valley detector **420** is a replica of the valley detector **410**, a DC level of $V.sub.b2$ will be approximately equal to $V'.sub.dc2$ and thus approximately equal to $V.sub.dc2$ when $V.sub.ip+$ and $V.sub.ip-$ are quiescent (i.e., $A(t)=0$). When $V.sub.ip+$ and $V.sub.ip-$ are dynamic (i.e., $A(t)+0$), an AC component of $V.sub.ip+$ and $V.sub.ip-$ will be coupled to $V'.sub.ip+$ and $V'.sub.ip-$, and a valley is detected and added to $V.sub.b2$ on top of $V.sub.dc2$, thus fulfilling the function described by equation (6).

(37) Note that a circuit principle of the P-type dynamic bias circuit **400** is the same as the N-type dynamic bias circuit **300**, except that peak detection is replaced by valley detection, $V.sub.dc1$ is replaced by $V.sub.dc2$, $V'.sub.dc1$ is replaced by $V'.sub.dc2$, and $V.sub.b1$ is replaced by $V.sub.b2$. Note that a NMOST is turned on when a gate voltage is high, while a PMOST is turned on when a gate voltage is low. That is why a NMOST can be used to detect a peak voltage, while a PMOST can be used to detect a valley voltage.

(38) Now reference is made to FIG. **1**. The N-type cascode amplifier **110** comprises NMOST **111a** and **111b** configured as a common-source amplifier, and NMOST **112a** and **112b** biased by $V.sub.b3$ and configured as a common-gate amplifier to establish a high output impedance and provide a high reverse isolation.

(39) P-type cascode amplifier **120** comprises PMOST **121a** and **121b** configured as a common-source amplifier, and PMOST **122a** and **122b** biased by $V.sub.b4$ and configured as a common-gate amplifier to establish a high output impedance and provide a high reverse isolation.

(40) A source of the N-type cascode amplifier **110** connects to a source of the P-type cascode amplifier **120** at a common source node **105**. This way, NMOST **111a** and PMOST **121b** form a first hybrid differential pair configured to receive and amplify a first hybrid differential signal comprising $V.sub.in+$ and $V.sub.ip-$, while NMOST **111b** and PMOST **121a** form a second hybrid differential pair configured to receive and amplify a second hybrid differential signal comprising $V.sub.in-$ and $V.sub.ip+$. Issues of ground bounce and source degeneration are mitigated, as explained in U.S. Pat. No. 10,447,218– B1. When the N-type cascode amplifier **110** and the P-type cascode amplifier **120** have approximately the same transconductance (i.e., voltage-to-current conversion ratio), and the first load **130** and the second load **140** have approximately the same impedance, a voltage $V.sub.CS$ at the common source node **105** is substantially stationary, an AC component of $V.sub.on+$ at node **101a** will be approximately equal to an AC component of

V.sub.op+ at node **102a**, while an AC component of V.sub.on- at node **101b** will be approximately equal to an AC component of V.sub.op- at node **102b**. In a further embodiment not shown in FIG. **1**, two additional capacitors are added, one placed across nodes **101a** and **102a** to enforce V.sub.in+ and V.sub.ip+ having an approximately equal AC component, and the other placed across nodes **101b** and **102b** to enforce V.sub.in- and V.sub.ip- having an approximately equal AC component. (41) When an amplitude of V.sub.i+ and V.sub.i- becomes larger, a peak of V.sub.in+ and V.sub.in- rises and a valley of V.sub.ip+ and V.sub.ip- falls; the N-type dynamic bias circuit **160** raises the first bias voltage V.sub.b1, causing a common-mode voltage of V.sub.in+ and V.sub.in- to rise and the N-type cascode amplifier **110** to be biased deeper into a class-A region; the P-type dynamic bias circuit **170** lowers the second bias voltage V.sub.b2, causing a common-mode voltage of V.sub.ip+ and V.sub.ip- to fall and the P-type cascode amplifier **120** to be biased deeper into a class-A region. This way, the N-type cascode amplifier **110** and the P-type cascode amplifier **120** can both be power efficient.

(42) The first load **130** comprises two inductors **131a** and **131b** and a capacitor **132** configured to form a resonant network to have a high impedance to allow the N-type cascode amplifier **110** to provide a high voltage gain, while providing a DC path to the power supply node V.sub.DD. The second load **140** comprises two inductors **141a** and **141b** and a capacitor **142** configured to form a resonant network to have a high impedance to allow the P-type cascode amplifier **120** to provide a high voltage gain, while providing a DC path to the ground node V.sub.SS. Note that capacitors **132** and **142** are optional, as a parasitic capacitance at nodes **101a**, **101b**, **102a**, and **102b** might be sufficient to form a resonance with inductors **131a**, **131b**, **141a**, and **141b**.

(43) In an alternative embodiment, capacitor **132** is placed across nodes **101a** and **102b** (instead of across nodes **101a** and **101b** as shown in FIG. **1**), while capacitor **142** is placed across nodes **102a** and **101b** (instead of across nodes **102a** and **102b** as shown in FIG. **1**). This alternative embodiment is workable because, as mentioned earlier, V.sub.in+ and V.sub.ip+ have approximately equal AC components, while V.sub.in- and V.sub.ip- have approximately equal AC components.

(44) In a further embodiment, amplifier **100** further comprises an output combiner **155** comprising four additional inductors **151a**, **151b**, **152a**, and **152b** configured to be magnetically coupled to inductors **131a**, **131b**, **141a**, and **141b** with coupling coefficients K.sub.na, K.sub.nb, K.sub.pa, and K.sub.pb, respectively, while the four additional inductors are connected in series and placed across an output node **109** and the ground node V.sub.SS. A combined output signal V.sub.o can be tapped from an output node **109**. A load circuit can be placed across the output node **109** and the ground node V.sub.SS.

(45) In the appended claims, the two capacitors **331** and **332** and the two resistors **333** and **334** are described as an AC coupling network that couples V.sub.in (comprising V.sub.in+ and V.sub.in-) to V'.sub.in (comprising V'.sub.in+ and V'.sub.in-) while setting a DC voltage of V'.sub.in (comprising V'.sub.in+ and V'.sub.in-) to V.sub.gn; NMOST **311** and NMOST **312** are referred to as a N-type common-mode source follower; and NMOST **321** and **322** are referred to as a replica N-type common-source follower. Likewise, the two capacitors **431** and **432** and the two resistors **433** and **434** are described as an AC coupling network that couples V.sub.ip (comprising V.sub.ip+ and V.sub.ip-) to V'.sub.ip (comprising V'.sub.ip+ and V'.sub.ip-) while setting a DC voltage of V'.sub.ip (comprising V'.sub.ip+ and V'.sub.ip-) to V.sub.gp; PMOST **411** and PMOST **412** are referred to as a P-type common-mode source follower; and PMOST **421** and **422** are referred to as a replica P-type common-source follower.

(46) In FIG. **1**, the trifilar transformer **150** is used to transform V.sub.i (comprising V.sub.i+ and V.sub.i-) to V.sub.in (comprising V.sub.in+ and V.sub.in-) and V.sub.ip (comprising V.sub.ip+ and V.sub.ip-). However, this is just an example but not limitation. In an alternative embodiment, two bifilar transformers are used: one for transforming V.sub.i (comprising V.sub.i+ and V.sub.i-) to V.sub.in (comprising V.sub.in+ and V.sub.in-), and the other for transforming V.sub.i (comprising V.sub.i+ and V.sub.i-) to V.sub.ip (comprising V.sub.ip+ and V.sub.ip-). This should be obvious to

those of ordinary skills in the art and thus not described in detail. Therefore, in the appended claims, a general term “transformer” is used without specifying a topology of the transformer; it is stated that the transformer comprises a primary coil, a first secondary coil, and a second secondary coil, and the statement is applicable to both the trifilar transformer **150** and the two bifilar transformers topologies.

(47) As shown in a flow diagram shown in FIG. 5, a method in accordance with an embodiment of the present disclosure comprises: (step **510**) receiving an input signal; (step **520**) transforming the input signal into a first transformed signal of a common-mode voltage equal to a first bias voltage; (step **530**) transforming the input signal into a second transformed signal of a common-mode voltage equal to a second bias voltage; (step **540**) detecting a peak of the first transformed signal and making the first bias voltage equal to a sum of a first DC (direct current) voltage and a first dynamic voltage approximately proportional to the peak; (step **550**) detecting a valley of the second transformed signal and making the second bias voltage equal to a sum of a second DC voltage and a second dynamic voltage approximately proportional to the valley; (step **560**) amplifying the first transformed signal into a first output signal across a first load using a N-type cascode amplifier; and (step **570**) amplifying the second transformed signal into a second output signal across a second load using a P-type cascode amplifier, wherein the N-type cascode amplifier and the P-type amplifier share a common source node.

(48) Those skilled in the art will readily observe that numerous modifications and alterations of the device and method may be made while retaining the teachings of the disclosure. Accordingly, the above disclosure should not be construed as limited only by the metes and bounds of the appended claims.

Claims

1. An amplifier comprising: a transformer configured to transform an input signal received from a primary coil into a first transformed signal of a first bias voltage and a second transformed signal of a second bias voltage via a first secondary coil and a secondary coil, respectively; a N-type dynamic bias circuit configured to receive the first transformed signal and output the first bias voltage in accordance with a first DC (direct current) voltage and a peak of the first transformed signal; a P-type dynamic bias circuit configured to receive the second transformed signal and output the second bias voltage in accordance with a second DC voltage and a valley of the second transformed signal; a N-type cascode amplifier configured to receive the first transformed signal and output a first output signal across a first load; and a P-type cascode amplifier configured to receive the second transformed signal and output a second output signal across a second load, wherein: the N-type cascode amplifier and the P-type cascode amplifier share a common source node.
2. The amplifier of claim 1, wherein the first bias voltage is provided to a center-tap of the first secondary coil, while the second bias voltage is provided to a center-tap of the second secondary coil.
3. The amplifier of claim 2, wherein the first bias voltage is equal to a sum of the first DC voltage and a dynamic voltage approximately proportional to a peak of the first transformed signal, while the second bias voltage is equal to a sum of the second DC voltage and a dynamic voltage approximately proportional to a valley of the second transformed signal.
4. The amplifier of claim 3, wherein: the N-type dynamic bias circuit comprises: an AC (alternating current) coupling network, a peak detector, a replica peak detector, and an operational amplifier, wherein: the AC coupling network couples the first transformed signal into a coupled signal of a DC voltage equal to a gate voltage, the peak detector comprises a N-type common-mode source follower configured to receive the coupled signal and output the first bias voltage, the replica peak detector comprises a replica N-type common-mode source follower configured to receive the gate

- voltage and output a replica DC voltage, and the operational amplifier outputs the gate voltage in accordance with a difference between the first DC voltage and the replica DC voltage.
5. The amplifier of claim 4, the operational amplifier and the replica N-type common-mode source follower form a negative feedback control loop to make the replica DC voltage approximately equal to the first DC voltage.
6. The amplifier of claim 3, wherein: the P-type dynamic bias circuit comprises: an AC (alternating current) coupling network, a valley detector, a replica valley detector, and an operational amplifier, wherein: the AC coupling network couples the second transformed signal into a coupled signal of a DC voltage equal to a gate voltage, the valley detector comprises a P-type common-mode source follower configured to receive the coupled signal and output the second bias voltage, the replica valley detector comprises a replica P-type common-mode source follower configured to receive the gate voltage and output a replica DC voltage, and the operational amplifier outputs the gate voltage in accordance with a difference between the second DC voltage and the replica DC voltage.
7. The amplifier of claim 6, the operational amplifier and the replica P-type common-mode source follower form a negative feedback control loop to make the replica DC voltage approximately equal to the second DC voltage.
8. The amplifier of claim 3, wherein: the N-type cascode amplifier comprises a cascade of a N-type common-source amplifier comprising a first NMOST (n-channel metal oxide semiconductor field effect transistor) and a N-type common-gate amplifier comprising a second NMOST, the first NMOST receiving the first transformed signal, and the second NMOST receiving a current output from the first NMOST and establishing the first output signal across the first load in accordance with a third bias voltage; and the P-type cascode amplifier comprises a cascade of a P-type common-source amplifier comprising a first PMOST (p-channel metal oxide semiconductor field effect transistor) and a P-type common-gate amplifier comprising a second PMOST, the first PMOST receiving the second transformed signal, and the second PMOST receiving a current output from the first PMOST and establishing the second output signal across the second load in accordance with a fourth bias voltage.
9. The amplifier of claim 3, wherein: the first load comprises a first inductor configured to provide a DC path to a power supply node and the second load comprises a second inductor configured to provide a DC path to a ground node.
10. The amplifier of claim 9 further comprising an output combiner comprising a serial connection of a third inductor magnetically coupled to the first inductor and a fourth inductor magnetically coupled to the second inductor.
11. A method comprising the following steps: receiving an input signal; transforming the input signal into a first transformed signal of a common-mode voltage equal to a first bias voltage; transforming the input signal into a second transformed signal of a common-mode voltage equal to a second bias voltage; detecting a peak of the first transformed signal and making the first bias voltage equal to a sum of a first DC (direct current) voltage and a first dynamic voltage approximately proportional to the peak; detecting a valley of the second transformed signal and making the second bias voltage equal to a sum of a second DC voltage and a second dynamic voltage approximately proportional to the valley; amplifying the first transformed signal into a first output signal across a first load using a N-type cascode amplifier; and amplifying the second transformed signal into a second output signal across a second load using a P-type cascode amplifier, wherein the N-type cascode amplifier and the P-type amplifier share a common source node.
12. The method of claim 11, wherein the step of “transforming the input signal into a first transformed signal of a common-mode voltage equal to a first bias voltage” comprises: using a magnetic coupling between a primary coil and a secondary coil to transform the input signal from the primary coil into the first transformed signal on the secondary coil of a center-tap connected to the first bias voltage.

13. The method of claim 11, wherein the step of “transforming the input signal into a second transformed signal of a common-mode voltage equal to a first bias voltage” comprises: using a magnetic coupling between a primary coil and a secondary coil to transform the input signal from the primary coil into the second transformed signal on the secondary coil of a center-tap connected to the second bias voltage.
14. The method of claim 11, wherein the step of “detecting a peak of the first transformed signal and making the first bias voltage equal to a sum of a first DC (direct current) voltage and a first dynamic voltage approximately proportional to the peak” comprises using a N-type dynamic bias circuit comprising: an AC (alternating current) coupling network, a peak detector, a replica peak detector, and an operational amplifier, wherein: the AC coupling network couples the first transformed signal into a coupled signal of a DC voltage equal to a gate voltage, the peak detector comprises a N-type common-mode source follower configured to receive the coupled signal and output the first bias voltage, the replica peak detector comprises a replica N-type common-mode source follower configured to receive the gate voltage and output a replica DC voltage, the operational amplifier outputs the gate voltage in accordance with a difference between the first DC voltage and the replica DC voltage.
15. The method of claim 11, wherein the step of “detecting a valley of the second transformed signal and making the second bias voltage equal to a sum of a second DC voltage and a second dynamic voltage approximately proportional to the valley” comprise using a P-type dynamic bias circuit comprising: an AC (alternating current) coupling network, a valley detector, a replica valley detector, and an operational amplifier, wherein: the AC coupling network couples the second transformed signal into a coupled signal of a DC voltage equal to a gate voltage, the valley detector comprises a P-type common-mode source follower configured to receive the coupled signal and output the second bias voltage, the replica valley detector comprises a replica P-type common-mode source follower configured to receive the gate voltage and output a replica DC voltage, the operational amplifier outputs the gate voltage in accordance with a difference between the second DC voltage and the replica DC voltage.
16. The method of claim 11, wherein the N-type cascode amplifier comprises a cascade of a N-type common-source amplifier comprising a first NMOST (n-channel metal oxide semiconductor field effect transistor) and a N-type common-gate amplifier comprising a second NMOST, the first NMOST receiving the first transformed signal, and the second NMOST receiving a current output from the first NMOST and establishing the first output signal across the first load in accordance with a third bias voltage; and the P-type cascode amplifier comprises a cascade of a P-type common-source amplifier comprising a first PMOST (p-channel metal oxide semiconductor field effect transistor) and a P-type common-gate amplifier comprising a second PMOST, the first PMOST receiving the second transformed signal, and the second PMOST receiving a current output from the first PMOST and establishing the second output signal across the second load in accordance with a fourth bias voltage.
17. The method of claim 16, wherein: the first load comprises a first inductor and the second load comprises a second inductor.
18. The method of claim 17, wherein the first inductor provides a DC path to a power supply node, while the second inductor provides a DC path to a ground node.
19. The method of claim 17 further comprising using an output combiner to combine the first output signal and the second output signal into a combined output signal.
20. The method of claim 19, wherein the output combiner comprises a serial connection of a third inductor magnetically coupled to the first inductor and a fourth inductor magnetically coupled to the second inductor.
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